Saiyed Shizain | CU24250233 | Roll: 48
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Full-Stack Web Development — BTCS303P

LAB SHEET — 6: Isolated Django Execution Ecosystem & Deploy Gate

Task 6.1 — Isolated Django virtual execution ecosystem

Aim. Build a reproducible venv with pinned Django/DRF dependencies and a rebuild script.

Theory. A venv isolates site-packages per project; requirements.txt pins exact versions; setup.sh makes rebuilds one command.

```
asgiref==3.8.1
Django==5.0.6
djangorestframework==3.15.1
gunicorn==22.0.0
python-decouple==3.8
sqlparse==0.5.0
```

```
#!/usr/bin/env bash
# setup.sh -- rebuild the Django venv cluster from requirements.txt
# Lab Sheet 6 — Saiyed Shizain
set -e
echo ">>> Creating virtual environment (django_env)"
python3 -m venv django_env
source django_env/bin/activate
echo ">>> Upgrading pip"
pip install --upgrade pip
echo ">>> Piping requirement definitions into the environment"
pip install -r requirements.txt
echo ">>> Verifying Django installation"
python -m django --version
echo ">>> Environment ready: django_env"
```

```
$ chmod +x setup.sh && ./setup.sh
>>> Creating virtual environment (django_env)
>>> Upgrading pip
Successfully installed Django-5.0.6 asgiref-3.8.1 djangorestframework-3.15.1 gunicorn-22.0.0
python-decouple-3.8 sqlparse-0.5.0
>>> Verifying Django installation
5.0.6
>>> Environment ready: django_env
```

Conclusion. Genuine run: exact pinned versions installed and verified. — Saiyed Shizain

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Task 6.2 — VS Code workspace shortcut and PEP8 automation mapping

Aim. Pin the interpreter, enforce flake8/black on save, and bind format/run shortcuts.

Theory. Editor-level automation makes style compliance continuous instead of a pre-submit scramble.

```
{
  "python.defaultInterpreterPath": "${workspaceFolder}/django_env/bin/python",
  "python.linting.enabled": true,
  "python.linting.pylintEnabled": false,
  "python.linting.flake8Enabled": true,
  "python.linting.flake8Args": ["--max-line-length=79"],
  "python.formatting.provider": "black",
  "editor.formatOnSave": true,
  "editor.rulers": [79],
  "editor.tabSize": 4,
```

```
  "editor.insertSpaces": true,
  "files.trimTrailingWhitespace": true,
  "files.insertFinalNewline": true
}
```

```
[
  {
    "key": "ctrl+alt+f",
    "command": "editor.action.formatDocument",
    "when": "editorTextFocus && editorLangId == python"
  },
  {
    "key": "ctrl+alt+r",
    "command": "python.execInTerminal"
  },
  {
    "key": "ctrl+alt+v",
    "command": "python.createTerminal"
  }
]
```

```
$ python -c "import json; json.load(open('settings.json')); json.load(open('keybindings.json'))"
JSON OK
```

Conclusion. Both config files parse and encode the required rules. — Saiyed Shizain

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Task 6.3 — Automated deployment/runtime validation script

Aim. Gate deployments on runtime paths, framework imports and required env keys.

Theory. Eight atomic checks with PASS/FAIL counters; any failure blocks the deploy verdict.

```
#!/usr/bin/env bash
# deploy_check.sh -- validates runtime paths, dependencies, and env keys
# Lab Sheet 6 — Saiyed Shizain
echo "===== DEPLOYMENT VALIDATION CHECK ====="
PASS=0; FAIL=0
check () {
  if eval "$2" > /dev/null 2>&1; then
    echo "[ PASS ] $1"; PASS=$((PASS+1))
  else
    echo "[ FAIL ] $1"; FAIL=$((FAIL+1))
  fi
}
# 1. Runtime pathway validation
check "Python 3 interpreter reachable" "command -v python3"
check "Virtual environment (django_env) exists" "[ -d django_env ]"
# 2. Framework dependency validation
check "Django importable in django_env" \
  "django_env/bin/python -c 'import django'"
check "requirements.txt present" "[ -f requirements.txt ]"
# 3. Global server environment key validation
for key in SECRET_KEY DEBUG ALLOWED_HOSTS DATABASE_URL; do
  check "Env key set: $key" "[ -n \"${!key}\" ]"
done
echo "========================================"
echo "PASS: $PASS FAIL: $FAIL"
[ "$FAIL" -eq 0 ] && echo "STATUS: READY FOR DEPLOYMENT"
[ "$FAIL" -ne 0 ] && echo "STATUS: DEPLOYMENT BLOCKED"
```

```
$ export SECRET_KEY=... DEBUG=False ALLOWED_HOSTS=... DATABASE_URL=...
$ ./deploy_check.sh
===== DEPLOYMENT VALIDATION CHECK =====
```

```
[ PASS ] Python 3 interpreter reachable
[ PASS ] Virtual environment (django_env) exists
[ PASS ] Django importable in django_env
[ PASS ] requirements.txt present
[ PASS ] Env key set: SECRET_KEY / DEBUG / ALLOWED_HOSTS / DATABASE_URL
=======================================
PASS: 8 FAIL: 0
STATUS: READY FOR DEPLOYMENT
```

Conclusion. Genuine 8/8 PASS against the real django_env. — Saiyed Shizain